

Priyansh Pathak

✉ pripat1008@gmail.com 📞 +91 9868782025
🌐 priyanshpathak.online 🐙 github.com/pripat1008 🔗 linkedin.com/in/pripat1008

RESEARCH SUMMARY

Research-oriented Computer Science undergraduate specializing in Machine Learning and Computer Vision. Experienced in deep learning-based face recognition, metric learning using Siamese neural networks and triplet loss, and video-based visual analysis. Seeking a summer research internship to contribute to ongoing work in visual perception and representation learning.

EDUCATION

SRM Institute of Science and Technology — B.Tech in Computer Science (AI & ML) *Aug 2023 – May 2027*
CGPA: **9.2 / 10**

Relevant Coursework: Machine Learning, Data Structures, Software Engineering, DBMS

IIT Ropar — Minor in Artificial Intelligence

Expected: 2025

TECHNICAL SKILLS

Programming: Python, Java, C++, SQL

Machine Learning: Deep Learning, Metric Learning, CNNs, Siamese Networks

Computer Vision: OpenCV, MTCNN, Face Recognition, Video Analysis

Frameworks & Tools: TensorFlow, Keras, MATLAB, Git

RESEARCH & INDUSTRIAL EXPERIENCE

Research Intern — IIT Mandi

May 2025 – Jul 2025

- Conducted research on a video-based attendance system using deep learning for face recognition.
- Designed and implemented a deep learning pipeline using TensorFlow/Keras with MTCNN for face detection and alignment.
- Trained Siamese neural networks using triplet loss for robust facial embedding learning.
- Performed dataset preprocessing, experimental evaluation, and comparative analysis with baseline methods.
- Conducted literature review of recent research in face recognition and metric learning.

AI Intern — AIEnsured

Jul 2025 – Sept 2025

- Developed and evaluated machine learning models with focus on robustness, fairness, and explainability.
- Automated model evaluation pipelines and documented experimental results.
- Assisted in integrating AI governance and evaluation tools into existing ML workflows.

SELECTED PROJECTS

Video-Based Face Recognition System Using Siamese Networks

Developed a Flask-based system for face recognition in classroom videos using Siamese neural networks, MTCNN for face detection, and triplet loss for metric learning. Automated preprocessing, model training, and attendance logging.

Image-to-Text Cultural Understanding Using Vision-Language Models

Built an AI application that generates culturally relevant narratives from user-uploaded images using vision-language models and structured JSON-based datasets, deployed via a Streamlit interface.

Real-Time Face Recognition Attendance System

Implemented a Python-based GUI application using Tkinter and OpenCV supporting secure training, real-time recognition, and CSV-based attendance management.

ADDITIONAL ACTIVITIES

Associate Lead — SRMKZILLA

2023 – Present

Contributor to community-led open-source and technical initiatives at SRM University.